R. ABHINAV

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PROFESSIONAL SUMMARY

Final-year B.Tech student in Artificial Intelligence and Data Science at Chennai Institute of Technology, with hands-on internship experience in backend development and end-to-end ML project delivery. Proven ability to build and deploy predictive models, optimise data pipelines, and contribute to production-grade systems. Seeking an internship to apply applied ML and software engineering skills to real-world, high-impact problems.

EDUCATION

CHENNAI INSTITUTE OF TECHNOLOGY, Chennai, India

Bachelor of Technology – Artificial Intelligence & Data Science | CGPA: 7.48 | Expected: May 2027

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Statistical Analysis, Database Management Systems

WORK EXPERIENCE

Froker – Software Development Engineer (SDE) Intern | 15 May 2025 – 1 July 2025

- Reduced API response latency by ~15% for a core service by collaborating with senior engineers to optimise data-processing routines, directly improving end-user experience.
- Identified and resolved 3 critical edge-case bugs during QA cycles, writing clean Python and Java code that adhered to team standards and prevented potential production failures.
- Delivered backend features across Agile sprints, consistently meeting sprint goals through daily stand-ups, code reviews, and proactive cross-functional communication.

PROJECTS

Predictive Maintenance System | Python, Scikit-learn, Random Forest, SVM, Pandas, Tableau

- Built a supervised ML pipeline to predict industrial machinery failures before they occur, targeting a reduction in unplanned downtime.
- Engineered features from sensor time-series data (vibration, temperature, RPM); handled class imbalance via SMOTE — achieved ~91% test accuracy.
- Visualised maintenance schedules and machine health KPIs in an interactive Tableau dashboard for non-technical stakeholders.
- Outcome: Simulated 23% reduction in maintenance cost and 17% improvement in equipment uptime vs. time-based baseline.

Pneumonia Detection using Deep Learning & Grad-CAM | TensorFlow, Keras, EfficientNetB0, OpenCV

- Developed a medical imaging classifier to detect pneumonia from chest X-rays using transfer learning (EfficientNetB0, ResNet50) with a two-phase training strategy.
- Applied data augmentation and preprocessing to improve generalisation; reduced overfitting on the validation set.
- Integrated Grad-CAM heatmaps via Gradient Tape to visually highlight infected lung regions — improving model interpretability for clinical review.

CERTIFICATIONS

- Android Developer Virtual Internship – AICTE & Google for Developers, India Edu Program | Oct – Dec 2024
- Introduction to Cybersecurity – Cisco Networking Academy | Sep 2023

TECHNICAL SKILLS

Programming:	Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), Java, SQL, HTML
Machine Learning:	Supervised Learning, Feature Engineering, Model Evaluation, EDA, Statistical Analysis, SMOTE, Transfer Learning
Visualization & Tools:	Tableau, Excel, Git, VS Code, TensorFlow, Keras, OpenCV
Languages:	English (Proficient), Tamil (Proficient)